Quantum Node Gravity (QNG): A Discrete Theory of Emergent Gravity, Time, and Structure

C.D Gabriel

February 9, 2026

Contents

Abstract **9**

1 Introduction **11**
1.1 Motivation and Background 11
1.2 Unresolved Problems in Physics 11
1.3 Objective of the QNG Framework 12
1.4 Empirical Basis: Pioneer, Lensing, Flyby 12
1.5 Structure of This Work 12

2 Ontological Foundations of QNG **15**
2.1 Absence of Classical Continuity 15
2.2 Discrete Nodes and Relational Space 15
2.3 Existence as Persistence Under Fluctuations 16
2.4 No Fundamental Forces, Only Stability 16
2.5 Time as Discrete Update Order 17
2.6 Formal Ontological Postulates of QNG 17

3 Fundamental Node Structure **19**
3.1 Definition of a Node 19
3.2 Discrete Volume (V_i) 19
3.3 Informational Memory (χ_i) — The Straton Field 20
3.4 Phase State (ϕ_i) 20
3.5 Connectivity and Emergent Geometry 20

4 Composite Time Mechanics (CTM) and the Straton **23**
4.1 The Redefinition of Time 23
4.2 Four Components of Composite Time 23
4.3 Straton as a Unit of Latent Reality 24
4.4 Decalajul Stratonic (ΔL) 24
4.5 Gravitational Memory and Dark Matter 24
4.6 Inertial Time vs. Observed Time 25
4.7 Illustration: Pioneer Anomaly Explained 25

5 Update Dynamics and the Stability Function Σ 27
5.1 The Evolution Rule $\mathcal{U}$ . . . 27
5.2 Role of Fluctuations (η) . . . 27
5.3 Definition of Stability Σ . . . 28
5.4 Selection Through Stability . . . 28
5.5 Stability Gradients and Emergent Forces . . . 28
5.6 Dynamic Persistence of Memory . . . 29
5.7 Implications for Dark Matter . . . 29

6 Emergence of Matter, Mass, and Inertia 31
6.1 From Nodes to Matter . . . 31
6.2 Mass as Temporal Stability . . . 31
6.3 Inertia as Informational Viscosity . . . 31
6.4 Emergence of Structure . . . 32
6.5 Relational Identity . . . 32
6.6 Quantum Phenomena as Stability Transitions . . . 33
6.7 Example: Stable Node Cluster as Particle . . . 33

7 Gravity and Dark Matter as Stability Gradient 35
7.1 Classical Gravity Revisited . . . 35
7.2 The Stability Field $\Sigma(x,t)$ . . . 35
7.3 Gravity as Gradient of Stability . . . 35
7.4 Time-Delayed Response . . . 36
7.5 Residual Acceleration from Lag . . . 36
7.6 Dark Matter Reinterpreted . . . 36
7.7 Predictions and Distinct Signatures . . . 37
7.8 Unified Perspective . . . 37

8 Experimental Signatures and Observational Tests 39
8.1 Scientific Testability . . . 39
8.2 Class A: Gravitational Lensing Discrepancies . . . 39
8.3 Class B: Flyby and Deep Space Anomalies . . . 40
8.4 Class C: Time-Dependent Dark Halos . . . 40
8.5 Class D: Lag-Corrected Newtonian Simulations . . . 40
8.6 Class E: Lab-Level Simulations (Quantum Graphs) . . . 41
8.7 Unified Signature Table . . . 41
8.8 Summary . . . 41

9 The Composite Time Model (CTM) and Pioneer-Type Anomalies 43
9.1 Rethinking Time: From Continuum to Composition . . . 43
9.2 The Role of the Straton (χ) . . . 43
9.3 Information Latency as Gravity Modifier . . . 44
9.4 Pioneer Anomaly Explained . . . 44
9.5 General Flyby Application . . . 44
9.6 Composite Time and GPS . . . 44

9.7 Formal Summary of Composite Time Model 45
9.8 Testable Predictions 45

10 Theoretical Implications and Comparison to Other Models 47
10.1 Motivation for a New Paradigm 47
10.2 Comparison to General Relativity (GR) 47
10.3 Comparison to Quantum Field Theory (QFT) 48
10.4 Comparison to String Theory 48
10.5 Comparison to Loop Quantum Gravity (LQG) 48
10.6 Implications for Unification 48
10.7 Philosophical Shift 49
10.8 Schematic Table: QNG vs Other Models 49

11 Straton Physics: Units, Measurement, and Applications 51
11.1 Definition of the Straton 51
11.2 Physical Meaning 51
11.3 Theoretical Role in QNG 52
11.4 Experimental Estimation 52
11.5 Straton-Based Equations 52
11.6 Comparison with Classical Units 52
11.7 Applications of the Straton 52
11.8 Future Research Directions 53

12 The Composite Time Geometry (CTG): Visualizing the Structure of Time and Causality 55
12.1 From Linearity to Structure 55
12.2 Time as Layered Dynamics 55
12.3 Topological View of Causality 56
12.4 Graphical Visualization (Suggested Diagrams) 56
12.5 Composite Light Cone 56
12.6 Quantum Interactions over Composite Time 56
12.7 Implications for Observables 57
12.8 Summary 57

13 The Dark Sector: Memory, Delay, and Emergent Matter 59
13.1 Redefining Dark Matter 59
13.2 Core Mechanism: Temporal Lag and Memory 59
13.3 Rewriting the Einstein Equation 60
13.4 Experimental Signatures 60
13.5 Formal Model: 60
13.6 Predictive Implications 61
13.7 What Dark Matter is Not 61
13.8 What Dark Matter Is (QNG View) 61
13.9 Cosmological Implications 61
13.10 Conclusion 61

14 Predictive Power: Experimental Tests and Observational Signatures 63
14.1 Scientific Falsifiability of QNG . . . 63
14.2 Prediction 1: Flyby Anomalies . . . 63
14.3 Prediction 2: Gravitational Lensing Distortions . . . 64
14.4 Prediction 3: Rotation Curves Evolution . . . 64
14.5 Prediction 4: Delayed Collapse Structures . . . 64
14.6 Prediction 5: Pulsar Timing Variations . . . 64
14.7 Test: Modified LIGO Signal Interpretation . . . 64
14.8 Lab-Scale Experiments (Future) . . . 65
14.9 Simulations and Numerical Models . . . 65
14.10 Conclusion: Scientific Strength of QNG . . . 65

15 Integration with Existing Physics: GR, QM, and QFT as Emergent Limits 67
15.1 Overview: Unification through Emergence . . . 67
15.2 Recovery of General Relativity (GR) . . . 67
15.3 Recovery of Quantum Mechanics (QM) . . . 68
15.4 Recovery of Quantum Field Theory (QFT) . . . 68
15.5 Integration Schema . . . 68
15.6 The Role of the Straton (χ) in All Three Domains . . . 69
15.7 Time Redefined across Frameworks . . . 69
15.8 Implications . . . 69
15.9 Conclusion . . . 69

16 Mathematical Core of QNG: Axioms, Structures, and Operators 71
16.1 Axiom Set of Quantum Node Gravity . . . 71
16.2 Node Structure . . . 71
16.3 Topological Graph Structure . . . 71
16.4 Temporal Update Rule . . . 72
16.5 Fluctuations and Stability . . . 72
16.6 Existence Condition . . . 72
16.7 Emergent Fields . . . 73
16.8 Memory Kernel (Dark Matter Mechanism) . . . 73
16.9 Composite Time Model (CTM) . . . 73
16.10 Stratonic Equation . . . 73
16.11 Advanced Operators (Optional) . . . 73
16.12 Conclusion . . . 74

17 Cosmological Extensions: QNG and the Large-Scale Universe 75
17.1 Initial Conditions and the Discrete Genesis . . . 75
17.2 Expansion as Network Proliferation . . . 75
17.3 Cosmic Microwave Background (CMB) . . . 75
17.4 Structure Formation . . . 76
17.5 Large-Scale Filamentary Web . . . 76
17.6 Cosmic Acceleration . . . 76
17.7 Predictions for Early Universe Anomalies . . . 76

17.8 Time Arrow and Entropy in QNG . . . 77
17.9 Inflation vs. Coherence Burst . . . 77
17.10 Cosmological Timeline in QNG . . . 77
17.11 Conclusion . . . 77

18 Applications and Engineering Speculations 79
18.1 Overview . . . 79
18.2 Straton-Based Mass Engineering . . . 79
18.3 Temporal Manipulation via τ . . . 79
18.4 Stratonic Lensing Systems . . . 80
18.5 QNG Resonators . . . 80
18.6 Information Transport Networks (QNG-Based) . . . 80
18.7 QNG-Driven Computation . . . 81
18.8 Experimental Suggestions . . . 81
18.9 Cautions and Limits . . . 81
18.10 Conclusion . . . 81

19 Testing, Falsifiability, and Open Predictions 83
19.1 Scientific Positioning of QNG . . . 83
19.2 Key Testable Components . . . 83
19.2.1 1. Stratonic Lag ($\Delta L = v \cdot \tau$) . . . 83
19.2.2 2. Stability Field Gradient ($\nabla\Sigma$) . . . 83
19.2.3 3. Memory Kernel (Historical Σ) . . . 84
19.3 Experimental Proposals . . . 84
19.3.1 A. Flyby Anomaly Tracking . . . 84
19.3.2 B. CMB Anomaly Analysis . . . 84
19.3.3 C. Simulated Nodal Networks . . . 84
19.3.4 D. Laboratory Analogues . . . 84
19.4 Falsifiability Pathways . . . 84
19.5 Predictions Unique to QNG . . . 85
19.6 Summary of Test Priorities . . . 85
19.7 Final Statement . . . 85

A Definitions, Symbols, Units, and Notations 87
A.1 Core Concepts and Entities . . . 87
A.2 Symbol Table . . . 88
A.3 Constants and Assumptions . . . 88
A.4 Notation Conventions . . . 88
A.5 Glossary of Terms . . . 88
A.6 Reference Units . . . 89

Abstract

Quantum Node Gravity (QNG) is a novel framework for quantum gravity that redefines spacetime as a discrete network of informational nodes governed by update rules, temporal persistence, and stability selection. In contrast to classical field-based theories, QNG proposes that geometry, mass, and gravity emerge from the dynamic evolution of a graph-structured universe, in which time is not a continuous coordinate but a discrete ordering of updates.

Each node carries a local volume, informational memory (χ), and phase, and interacts with its neighbors through stochastic yet regulated transformations. The core mechanism driving the system is the *Stability Function* (Σ), which governs existence: only configurations that remain coherent under fluctuations persist over time.

A central contribution of QNG is the introduction of the *Straton* ($\chi = m/c$), a new unit representing the latent informational load of matter within the spacetime network. This leads to a *Composite Time Model* (CTM), in which observable time includes delay effects arising from the processing of straton fluxes—generating small but potentially detectable anomalies in gravitational systems, such as those reported in Pioneer and planetary flyby data.

Within this framework, **gravity is not a fundamental force**, but rather the gradient of historical informational stability. Phenomena commonly attributed to dark matter are reinterpreted as memory effects in the gravitational field. QNG thus provides a coherent, discrete, and testable model of spacetime, matter, and gravitation, compatible with current astrophysical observations and open to future experimental validation.

Chapter 1

Introduction

1.1 Motivation and Background

The pursuit of a quantum theory of gravity has remained one of the most formidable challenges in theoretical physics. While General Relativity (GR) successfully describes gravity as the curvature of spacetime, and Quantum Mechanics (QM) governs phenomena at the smallest scales, the two frameworks are fundamentally incompatible in extreme regimes—such as black hole singularities and the early universe.

Furthermore, discrepancies between gravitational effects observed on cosmic scales and the distribution of visible matter (e.g., galactic rotation curves, gravitational lensing, and cosmic structure formation) have led to the inference of dark matter, a mysterious substance that interacts gravitationally but not electromagnetically. Despite decades of experimental searches, no dark matter particle has been detected, and its nature remains unknown.

1.2 Unresolved Problems in Physics

QNG emerges from a reassessment of foundational assumptions in modern physics. Rather than beginning with classical continua or force fields, QNG is built on discrete information units and interaction rules. This framework seeks to address the following open issues:

- **Quantum Gravity:** No existing theory fully reconciles quantum theory with spacetime dynamics.
- **Dark Matter:** Observations require additional gravitational effects that are not accounted for by visible matter.
- **Anomalies:** Space missions like Pioneer and certain flyby trajectories exhibit deviations from expected gravitational behavior.
- **Nature of Mass:** The origin of inertia and the precise nature of mass remain conceptually incomplete.
- **Continuum Breakdown:** At Planck-scale regimes, spacetime may no longer be continuous.

1.3 Objective of the QNG Framework

The core hypothesis of QNG is that spacetime is a discrete, informational network of nodes, and that mass, gravity, and even dark matter are emergent phenomena arising from temporal stability in this network. This leads to a redefinition of key physical concepts:

- **Mass** is reinterpreted as temporal stability (Σ_{temp}).
- **Gravity** is the spatial gradient of the network's stability field ($\nabla\Sigma$).
- **Time** is not a continuous dimension but an irreversible ordering of updates.
- **Dark matter** arises from delayed gravitational memory, not exotic particles.

1.4 Empirical Basis: Pioneer, Lensing, Flyby

The framework is motivated by a set of unresolved observational phenomena, including:

- The **Pioneer anomaly**, which suggested a small unexplained acceleration in spacecraft tracking data.
- **Gravitational lensing offsets** that exceed expectations based on luminous mass alone.
- **Flyby anomalies**, where spacecraft gain or lose energy in ways that are not predicted by standard gravitational models.

These anomalies suggest that gravity may not respond instantaneously to mass, but rather evolves with a memory effect. QNG captures this through the concept of *stratonic lag*, a measurable delay introduced by information processing constraints in spacetime itself.

1.5 Structure of This Work

This paper presents Quantum Node Gravity in its complete formal architecture. The chapters are organized as follows:

- **Chapter 2:** Introduces the ontological foundations — nodes, relations, discrete time, and stability.
- **Chapter 3:** Defines the internal structure of each node and emergent spatial properties.
- **Chapter 4:** Details the Composite Time Model and introduces the Straton.
- **Chapter 5:** Presents the update dynamics and stability-based existence condition.
- **Chapter 6:** Describes matter and mass as emergent from persistent node configurations.

- **Chapter 7:** Derives gravity from the stability gradient.
- **Chapter 8:** Offers a redefinition of dark matter as persistent gravitational memory.
- **Chapter 9:** Expands on the Straton and its physical implications.
- **Chapter 10:** Provides simulations, visualizations, and lensing diagrams.
- **Chapter 11:** Analyzes observational data for validation and prediction.
- **Chapter 12:** Discusses metaphysical implications and computational parallels.
- **Chapter 13:** Summarizes findings and outlines future research.

Chapter 2

Ontological Foundations of QNG

2.1 Absence of Classical Continuity

One of the key philosophical departures of Quantum Node Gravity (QNG) is the rejection of spacetime as a continuous manifold. Traditional physics assumes a smooth background upon which fields and particles are defined. However, quantum gravity candidates and Planck-scale reasoning suggest that continuity may be an emergent illusion. QNG adopts a discrete ontology from the outset: the universe is made of nodes, not points in a smooth continuum.

This rejection is motivated by:

- The non-renormalizability of gravity when treated as a quantum field on a continuous spacetime.
- The appearance of minimal length scales in black hole thermodynamics and loop quantum gravity.
- The holographic principle, which implies a finite information density in bounded regions.

QNG posits that all physical entities emerge from the structure and evolution of a discrete, relational network. Space, time, mass, and curvature are derived—not assumed.

2.2 Discrete Nodes and Relational Space

The fundamental building block in QNG is the **informational node**. There is no absolute space in which these nodes are embedded. Instead, they are connected via an abstract graph $G = (N, E)$, where:

- N: the set of nodes $\mathcal{N}_i$,
- E: the set of edges representing direct relational links.

Distance and **geometry** are not primitive but emerge from the structure of these connections and the internal states of the nodes. This framework aligns with relational theories of spacetime, where relations—not entities—are fundamental.

The internal state of a node is given by:

$$\mathcal{N}_i = (V_i, \chi_i, \phi_i)$$

where:

- V_i: discrete volumetric assignment,
- χ_i: memory or informational content (Straton units),
- ϕ_i: phase or quantized interaction state.

2.3 Existence as Persistence Under Fluctuations

Unlike traditional physics where existence is assumed axiomatically, QNG defines existence dynamically. A node or structure is said to *exist* if and only if it **persists through time** despite fluctuations.

$$\text{Existence} \iff \Sigma_i \geq \Sigma_{\min}$$

Here, Σ_i is the node's stability function, and $\Sigma_{\min}$ is a universal threshold. This principle introduces a natural selection mechanism into the physics: only stable, persistent configurations survive the update dynamics.

This aligns with observable reality, where coherent particles (e.g., protons, atoms, stars) endure while transient fluctuations vanish.

2.4 No Fundamental Forces, Only Stability

QNG eliminates the need for fundamental forces as independent fields. Instead, interactions arise from the tension and equilibrium between neighboring node states. Gravitation, for instance, is not a pull or field but the result of a gradient in *informational stability* across the network:

$$\vec{g} = -\nabla\Sigma$$

In this view:

- Inertia = resistance to altering stable informational states.
- Gravitation = propagation of destabilization across a node network.
- Dark matter = accumulated stability lag from prior configurations.

Key Argument: If the fundamental fabric of spacetime is information, then the only meaningful physical quantities are those that affect or resist change to that information. Hence, forces become derivative concepts.

2.5 Time as Discrete Update Order

Time in QNG is not a geometric coordinate but a **discrete ordering of updates**. Each node updates according to a local or global rule $\mathcal{U}$, influenced by neighboring states and a noise term η_i. This produces a causal structure that resembles a directed acyclic graph (DAG).

$$T \equiv \text{update sequence} \Rightarrow t_n \in \mathbb{N}$$

There is no absolute flow or continuum of time—only a record of changes and the order in which they occur. This matches well with computational models, cellular automata, and theories that embed physics in computation.

Implications

- **Arrow of Time**: Emerges naturally as the direction of persistent updates.
- **No Time Travel**: Impossible since updates are non-invertible.
- **Causality**: Imposed by update rules and graph structure.

2.6 Formal Ontological Postulates of QNG

Postulate 1: Discreteness — The universe is composed of discrete informational nodes, not continuous fields.

Postulate 2: Relationality — Space is the outcome of connections among nodes, not a background.

Postulate 3: Temporal Order — Time is the sequence of updates, not a dimension.

Postulate 4: Stability — Physical persistence is determined by resistance to fluctuation.

Postulate 5: Emergence — Matter, mass, and curvature arise from the evolution of node states.

Postulate 6: No Absolute Entities — No absolute space, time, or force exists independently of node behavior.

These postulates position QNG as a radical but coherent attempt to unify quantum mechanics, gravity, and information theory under a new ontological paradigm.

Chapter 3

Fundamental Node Structure

3.1 Definition of a Node

In the Quantum Node Gravity (QNG) framework, a node is the irreducible informational unit of reality. It is not a point in space, but a discrete system possessing state, volume, memory, and interaction phase. The evolution of the universe is understood as the collective update of this network of nodes.

Each node $\mathcal{N}_i$ is defined formally as:

$$\mathcal{N}_i = (V_i, \chi_i, \phi_i)$$

where:

- V_i is the assigned discrete volume of the node,
- χ_i is the informational memory — a measure of accumulated or encoded interactions,
- ϕ_i is the interaction phase — a local state related to synchronization and quantization.

3.2 Discrete Volume (V_i)

Unlike continuum physics, QNG defines volume in fundamental, non-divisible units. Each node represents a "quantum of space," and the macroscopic volume of objects or regions arises from aggregating these nodes.

Properties of V_i:

- Quantized: $V_i \in \mathbb{N} \cdot V_0$, where V_0 is the fundamental unit.
- Non-continuous: There is no interpolation between nodes.
- Locally finite: Prevents ultraviolet divergences common in field theories.

The assumption of a minimum volume aligns with Planck-scale considerations and the holographic bound.

3.3 Informational Memory (χ_i) — The Straton Field

χ_i is the informational content or "load" a node carries. It is measured in units called **Stratons**, defined as:

$$\chi = \frac{m}{c}$$

This reflects the amount of "inertia" the node contributes to the network — a mass-equivalent latency that must be processed over time.

Interpretation:

- χ represents latent informational reality.
- Higher χ_i means more resistance to state change.
- It contributes directly to the temporal delay of reconfiguration (τ).

Stratons generalize the concept of mass as an informational burden distributed across the spacetime graph.

3.4 Phase State (ϕ_i)

ϕ_i encodes the dynamic interaction potential of the node. It governs how nodes couple, synchronize, and exchange state information.

Role of ϕ_i:

- Encodes quantization rules (e.g., allowed transitions).
- Mediates coherence and interference phenomena.
- Potentially links to electromagnetic and gauge-like interactions.

Nodes with synchronized ϕ_i values form coherent clusters — the basis of emergent particles and fields.

3.5 Connectivity and Emergent Geometry

Nodes are not embedded in space. Instead, they are linked via a dynamic graph $G = (N, E)$. Geometry emerges from the structure and evolution of this graph:

- Distance is defined by minimal path length on the graph.
- Curvature is encoded in topological deviations (e.g., node density gradients).
- Time emerges from the evolution (ordering) of the network.

This approach reflects a shift from geometry-first to **relational-first** physics.

Summary Equation (Node Object):

$$\boxed{\mathcal{N}_i(t) = \left(V_i, \frac{m_i}{c}, \phi_i(t)\right)} \Rightarrow \text{Updated via } \mathcal{U}$$

Interpretation: Each node evolves according to update rules $\mathcal{U}$, influenced by its own state, neighbor states, and noise. Through time, coherent structures stabilize, forming matter and observable fields.

Chapter 4

Composite Time Mechanics (CTM) and the Straton

4.1 The Redefinition of Time

In traditional physics, time is treated either as a continuous geometric coordinate (in General Relativity) or as a parameter of unitary evolution (in Quantum Mechanics). QNG rejects both views and instead defines time as a composite process, built from discrete causal updates in a finite-information network.

Time is not an absolute backdrop, but an emergent result of how nodes in the universe process and transmit information.

4.2 Four Components of Composite Time

The observable time (T_C) experienced by physical systems is composed of multiple layers:

1. **Reference Time** (t_R) – The intrinsic refresh rate of the universe's node network. It defines the "tick" of the fundamental system and cannot be slowed or accelerated.
   Metaphor: The processor clock speed in a computer.

2. **Relaxation Time** (τ) – The delay introduced by the system's need to redistribute the informational load (χ) from one node to another.
   Metaphor: Latency in a distributed system.

3. **Straton Load** ($\chi = m/c$) – The quantity of informational mass a node carries and which requires transfer in order to evolve.

4. **Stability Gradient** ($\nabla\Sigma$) – The change in system stability due to past updates, acting as a memory field that influences dynamics.

Together, they form the **Composite Time**:

$$T_C = t_R + \tau(\chi) + \nabla\Sigma \cdot \tau$$

This expression reveals that time is not absolute, but a response of the system to its own inertia and history.

4.3 Straton as a Unit of Latent Reality

The **Straton** is defined as:

$$\chi = \frac{m}{c}$$

It quantifies how much of a system's informational weight needs to be propagated through the network at each step. Unlike classical mass, χ directly determines the time lag in system response.

Properties of the Straton:

- Units: [kg·s/m] — not traditional energy or mass.
- Interpretation: The "shadow" mass leaves in the temporal structure of the universe.
- Physical role: Governs the processing cost of motion or interaction.

In the QNG framework, gravity does not react instantly to mass, but to the flow of Stratons across the network. This leads to the appearance of inertial drag, gravitational memory, and emergent lensing effects.

4.4 Decalajul Stratonic (ΔL)

When a system with high χ moves quickly, it gets ahead of its own stability field — a phenomenon called the **Stratonic Lag**. The spatial manifestation of this lag is:

$$\Delta L = v \cdot \tau$$

Implications:

- Predicts residual gravitational forces.
- Explains anomalies like Pioneer and flyby effects.
- Replaces the need for exotic particles to explain excess gravity.

4.5 Gravitational Memory and Dark Matter

Stratonic lag accumulates in the stability field. This field $\Sigma(x, t)$ does not instantly reflect the current position of mass but lags behind, creating regions of excess gravitational pull — observed as dark matter:

$$\text{Dark Matter} \approx \Sigma_{\text{historical}} - \Sigma_{\text{instantaneous}}$$

This perspective transforms dark matter from a particle mystery into a coherent, testable memory effect caused by informational processing constraints.

4.6 Inertial Time vs. Observed Time

The CTM also explains inertial effects:

- Bodies with higher χ require more network resources to change motion — hence, they exhibit greater inertia. - Fast-changing systems experience greater lag, leading to relativistic effects naturally.

This offers a reinterpretation of special relativity:

$$\text{Time dilation} \Rightarrow \text{Delay in information processing, not geometry.}$$

4.7 Illustration: Pioneer Anomaly Explained

Consider a spacecraft with velocity v, mass m, and known trajectory. Classical GR predicts its position using instantaneous spacetime curvature.

In QNG:

- The spacecraft carries $\chi = m/c$,
- It propagates through nodes with relaxation time τ,
- The resulting gravitational pull from the stability field is offset by $\Delta L = v \cdot \tau$,
- The observed anomaly is the difference between actual and lagged acceleration.

This matches the flyby and Pioneer anomalies to first order, with no need for dark matter or modification of GR equations — only an extension into informational dynamics.

Chapter 5

Update Dynamics and the Stability Function Σ

5.1 The Evolution Rule $\mathcal{U}$

In QNG, the universe evolves via discrete update steps. Each node $\mathcal{N}_i$ updates based on its current state, the states of its neighbors, and a stochastic fluctuation term. The general update rule is:

$$\mathcal{N}_i(t+1) = \mathcal{U}\big(\mathcal{N}_i(t), \{\mathcal{N}_j(t)\}_{j\in\text{Neighbors}(i)}, \eta_i(t)\big)$$

Where:

- $\mathcal{U}$: The universal update operator (local but possibly nonlinear),
- η_i: Random fluctuation (noise term),
- $\{\mathcal{N}_j\}$: Neighboring nodes connected in the graph.

This rule governs all physical evolution: from vacuum stability to the formation of particles and gravitational fields.

5.2 Role of Fluctuations (η)

Fluctuations are not errors but essential components of the QNG system. The physical universe is inherently noisy at the Planck scale.

Why η matters:

- It drives exploration of state space.
- It breaks perfect symmetry — enabling structure formation.
- It introduces quantum-like uncertainty at the fundamental level.

Fluctuations can be modeled as Gaussian noise, bounded by local informational thresholds, or encoded in complex phase behavior (ϕ_i).

5.3 Definition of Stability Σ

Each node carries a stability score $\Sigma_i \in [0, 1]$, determining whether it persists into the next update. The general form is:

$$\Sigma_i = \Sigma_\chi \cdot \Sigma_{\text{struct}} \cdot \Sigma_{\text{temp}} \cdot \Sigma_\phi$$

Where:

- Σ_χ: informational consistency (based on χ_i),
- Σ_{struct}: topological robustness (based on neighborhood),
- Σ_{temp}: temporal persistence under change,
- Σ_ϕ: coherence of phase with neighbors.

If $\Sigma_i < \Sigma_{\min}$, the node **ceases to exist** — it vanishes from the graph.

Key Postulate:

$$\boxed{\text{Existence} \iff \Sigma_i \geq \Sigma_{\min}}$$

5.4 Selection Through Stability

Unlike theories where physical entities exist by assumption (particles, spacetime), QNG imposes a form of natural selection:

- Unstable nodes decay.
- Clusters of high-Σ nodes persist and interact.
- Physical objects (e.g. matter, stars, black holes) are emergent attractors in the space of stable structures.

This principle provides a deep unification of ontology, evolution, and physics.

5.5 Stability Gradients and Emergent Forces

The gradient of the stability field across nodes defines what we interpret as physical force fields:

$$\vec{F}_{\text{grav}} = -\nabla\Sigma$$

This result appears at large scale as curvature in General Relativity, but in QNG it arises from microscopic informational tension. Inertia and gravitation become two sides of the same informational coin.

5.6 Dynamic Persistence of Memory

When a node moves or changes, its previous presence is still encoded in the surrounding stability field. This memory generates residual forces and explains dark matter effects:

$$\Sigma(x,t) = \int_{-\infty}^{t} K(t-t') \cdot \chi(x,t')\, dt'$$

Where K is a memory kernel (e.g. exponential decay with timescale τ).

Key Result:

$$\text{Residual Acceleration} \approx -\tau \cdot (\vec{v} \cdot \nabla)\nabla\Sigma$$

This term explains:

- Flyby anomalies,
- Lensing offset in galaxy clusters,
- Apparent excess mass in stable but matter-free regions.

5.7 Implications for Dark Matter

The field Σ carries the memory of past mass distributions. Thus:

$$\text{Dark Matter} = \Sigma_{\text{past}} - \Sigma_{\text{now}}$$

No new particles are needed — only memory dynamics of the gravitational stability field. This model predicts time-dependent dark matter halos and their dissipation over long timescales, a unique testable prediction of QNG.

Chapter 6

Emergence of Matter, Mass, and Inertia

6.1 From Nodes to Matter

In QNG, **matter** is not a substance, but a stable configuration of nodes. When a group of nodes forms a persistent, self-reinforcing pattern under the update rule $\mathcal{U}$, it becomes what we observe as a physical object.

Matter is defined as:

$$\text{Matter} \equiv \text{Stable Attractor in the Node Update Network}$$

These attractors resist decay from fluctuations and propagate as coherent structures through the graph. Their persistence over time defines their existence.

6.2 Mass as Temporal Stability

In classical physics, mass is treated as intrinsic. In QNG, it emerges from how long a node structure can persist despite fluctuations.

$$m \propto \Sigma_{\text{temp}}$$

Interpretation:

- High-mass structures = high temporal stability.
- Low-mass = easily disturbed and short-lived.
- *Mass is not "how much stuff", but "how long a pattern survives".*

6.3 Inertia as Informational Viscosity

In QNG, motion means propagating the informational load (χ) through the network. This propagation has cost: the system resists it, and this resistance is what we perceive as **inertia**.

$$\text{Inertia} \propto \text{Informational Viscosity} \propto \chi \cdot \tau$$

Implications:

- Fast-moving, high-χ systems encounter lag.
- Inertial frames arise from balancing informational fluxes.
- Inertia is not fundamental — it is a network processing constraint.

6.4 Emergence of Structure

Complex matter — atoms, particles, macroscopic objects — emerges when local attractors link to form larger, stable aggregates.

Conditions for Emergence:

1. Local stability in all four Σ components,
2. Connectivity: ability to exchange information with neighbors,
3. Synchronization of phase: coherence in ϕ_i.

This naturally reproduces phenomena such as:

- Bound states (e.g. particles),
- Composite objects (e.g. molecules, solids),
- Propagation of excitations (e.g. waves, fields).

6.5 Relational Identity

A structure is not defined by fixed position, but by pattern:

$$\text{Identity} = \text{Persistent Configuration of Update Relations}$$

Key Principle: *There is no "thing-in-itself", only processes that endure.*

This aligns with quantum ideas of indistinguishability and with relativity's block-universe constraints — without assuming spacetime as continuous.

6.6 Quantum Phenomena as Stability Transitions

QNG predicts that quantum effects such as:

- Superposition,
- Tunneling,
- Decoherence,

arise naturally when a system is near the edge of stability:

$$\Sigma \approx \Sigma_{\min} \Rightarrow \text{Uncertain Existence}$$

This leads to probabilistic outcomes, not because nature is indeterminate, but because stability is on the threshold — just like a particle "choosing" to decay.

6.7 Example: Stable Node Cluster as Particle

Imagine a 5-node configuration where:

- All nodes satisfy $\Sigma_i > \Sigma_{\min}$,
- They exchange ϕ_i in a loop,
- Total χ is conserved across updates.

This configuration can persist over many updates and propagate through the network. It behaves like a particle — and different patterns correspond to different types (mass, spin, etc.).

Chapter 7

Gravity and Dark Matter as Stability Gradient

7.1 Classical Gravity Revisited

In General Relativity (GR), gravity is described as spacetime curvature generated by mass-energy. However, this assumes a continuous spacetime manifold and an instantaneous adjustment of geometry.

QNG reinterprets gravity not as curvature of space, but as a response to gradients in the informational stability field Σ.

7.2 The Stability Field $\Sigma(x, t)$

In QNG, each point in the network carries a scalar field Σ, representing the cumulative resistance to informational change — the "memory tension" of the system.

$$\Sigma(x, t) = \int_{-\infty}^{t} K(t - t') \cdot \chi(x, t')\, dt'$$

Where:

- $K(\Delta t)$ is the relaxation kernel, e.g. $K(\Delta t) = \frac{1}{\tau} e^{-\Delta t/\tau}$,
- $\chi(x, t')$ is the local straton flux.

7.3 Gravity as Gradient of Stability

The gravitational "force" emerges from spatial differences in this stability field:

$$\vec{g}(x, t) = -\nabla \Sigma(x, t)$$

This vector field directs motion toward regions of higher persistence (stability), mimicking the attractive nature of gravity.

Implications:

- Gravity is not a field sourced by mass, but by **information flux memory**.
- Gravitational lensing and orbital dynamics emerge from field gradients, not geometry.
- No singularities appear — Σ is always bounded and continuous.

7.4 Time-Delayed Response

Unlike GR, QNG introduces a response delay via relaxation time τ:

$$\nabla\Sigma(t) \approx \nabla\Sigma(t-\tau)$$

This delay causes spatial offset between mass and its gravitational effect, leading to anomalies in highly dynamic systems.

7.5 Residual Acceleration from Lag

The lag between matter and stability field yields a residual acceleration:

$$\vec{a}_{\text{res}} \approx -\tau \cdot (\vec{v} \cdot \nabla)\nabla\Sigma$$

Predicted effects:

- Non-Newtonian accelerations during planetary flybys,
- Forward-shifted gravitational lensing in clusters,
- Small deviations in deep space probes (e.g. Pioneer anomaly).

7.6 Dark Matter Reinterpreted

Standard cosmology postulates dark matter to explain gravitational discrepancies.

In QNG, these discrepancies are due to the delayed response of the Σ field to past mass configurations:

$$\text{Dark Matter} \equiv \Sigma_{\text{historical}} - \Sigma_{\text{instantaneous}}$$

Observational Reinterpretation:

- Lensing maps = $\nabla\Sigma$, not mass.
- Offset in cluster mergers = relaxation time.
- No need for WIMPs, axions, or supersymmetric particles.

7.7 Predictions and Distinct Signatures

QNG predicts:

1. **History-dependent gravity** — dark matter halos vary with past dynamics.
2. **Directional anomalies** — acceleration depends on $\vec{v}$.
3. **Dark matter can dissipate** — via long-term field relaxation.

These effects are *not* predicted by GR or particle dark matter theories, and offer a clear experimental path forward.

7.8 Unified Perspective

Final formula:

$$\vec{g}(x,t) = -\nabla \left(\int_{-\infty}^{t} K(t-t') \cdot \chi(x,t')\, dt' \right)$$

This single equation accounts for:

- Classical gravity (as $\tau \to 0$),
- Dark matter (as memory effects),
- Residual anomalies (as directional lag).

Chapter 8

Experimental Signatures and Observational Tests

8.1 Scientific Testability

Quantum Node Gravity is a falsifiable theory. It predicts deviations from General Relativity (GR) and standard cosmology that can be observed in current or near-future data. These deviations are not arbitrary but structurally required by the memory and update dynamics of the Σ field.

8.2 Class A: Gravitational Lensing Discrepancies

Prediction:

Gravitational lensing will reflect the stability field $\nabla\Sigma$, not the instantaneous matter distribution.

Implications:

- Offset between baryonic mass and lensing center.
- Asymmetric halos depending on galactic history.
- Memory signatures in post-merger clusters (e.g., Bullet Cluster).

Available Data:

Hubble, JWST, and Euclid provide direct mappings of lensing fields, usable to compare Σ-driven lensing vs. classical mass-based predictions.

8.3 Class B: Flyby and Deep Space Anomalies

Prediction:

Probes experiencing changes in velocity direction will show residual accelerations consistent with:

$$\vec{a}_{\text{res}} \approx -\tau \cdot (\vec{v} \cdot \nabla)\nabla\Sigma$$

Historical Observations:

- Flyby anomaly (Galileo, NEAR, Rosetta, Cassini).
- Pioneer anomaly (before thermal re-analysis).

QNG suggests that these are not mere instrument artifacts, but real directional lags in the Σ field.

8.4 Class C: Time-Dependent Dark Halos

Prediction:

Dark matter halos will evolve over time even in the absence of new matter, due to slow relaxation of the stability field.

Testable Effects:

- Galaxies with similar visible mass but different formation histories will have different lensing profiles.
- Halos may appear to "fade" in very old, isolated systems.

8.5 Class D: Lag-Corrected Newtonian Simulations

Standard N-body simulations assume instantaneous gravity. By incorporating a memory kernel (with relaxation time τ), simulations will produce QNG-compliant structures.

$$\Sigma(t) = \int_{-\infty}^{t} K(t - t') \cdot \chi(t')\, dt'$$

Expected Output:

- Enhanced coherence of galactic rotation curves.
- Offset clustering effects in early universe models.
- Dissipation of excess gravitational potential over long epochs.

8.6 Class E: Lab-Level Simulations (Quantum Graphs)

Even though the Planck-scale QNG regime is unreachable, we can simulate QNG-like node networks using:

- Quantum cellular automata.
- Graph neural networks with memory persistence.
- Phase-coupled oscillator networks.

These simulations can reproduce:

- Node stability,
- Information lag,
- Emergent attractors.

8.7 Unified Signature Table

Phenomenon	**GR / DM Prediction**	**QNG Prediction**
Flyby anomaly	None	Residual directional lag
Lensing offset	Particle dark matter	Stability field gradient
Dark halo fading	Not predicted	Field memory dissipation
Rotation curves	Need exotic mass	Emergent from Σ
Bullet cluster offset	Requires new particle	Delayed Σ alignment

Table 8.1: Comparative predictions: GR + dark matter vs QNG

8.8 Summary

QNG makes strong, coherent, and testable predictions that differ sharply from GR + particle dark matter. The presence of a relaxation time τ and the Σ memory field offer a new interpretation of longstanding anomalies and open new paths for empirical validation.

Chapter 9

The Composite Time Model (CTM) and Pioneer-Type Anomalies

9.1 Rethinking Time: From Continuum to Composition

In QNG, time is not a universal continuous parameter but a composite effect of multiple layers:

1. **Reference Time** (t_R) – the update clock of the node network,
2. **Relaxation Time** (τ) – the memory delay required to propagate information,
3. **Observed Time** (T_C) – the macroscopic time we measure.

Key Concept:

$$T_C = t_R + \Delta L(\chi, v)$$

Where:

$$\Delta L = v \cdot \tau(\chi)$$

This lag term represents the information latency from moving Straton content across the network.

9.2 The Role of the Straton (χ)

We define:

$$\chi = \frac{m}{c}$$

Interpretation:

- χ is the latent informational pressure exerted by a mass m on the spacetime network.
- High χ implies greater delay in being "processed" into the next nodal update.
- χ is a measurable, theoretical unit — a candidate for a new physical constant.

9.3 Information Latency as Gravity Modifier

Each node must process the incoming Straton flux from neighboring updates. The larger the χ, the longer this takes. This delay creates a spatial-temporal offset, which at macroscopic scale appears as anomalous acceleration.

Formal Equation:

$$\vec{a}_{\text{lag}} \propto -\chi \cdot \nabla \left(\frac{d\Sigma}{dt} \right)$$

This acceleration term emerges only when motion occurs through dynamically changing Σ fields.

9.4 Pioneer Anomaly Explained

- The Pioneer spacecraft exhibited an unexplained sunward acceleration.
- Classical physics failed to explain this, later attributing it to thermal radiation.
- QNG proposes: this is the result of a lagging Σ-field caused by forward movement of χ-heavy probes.

$$\vec{a}_{\text{Pioneer}} \approx -\tau \cdot (\vec{v} \cdot \nabla) \nabla \Sigma$$

Interpretation: The probe is ahead of its own gravitational "footprint" in the stability field, creating a pull backward.

9.5 General Flyby Application

QNG's lag model predicts similar effects in flyby scenarios:

- Directional anomaly: occurs only when velocity vector shifts relative to $\nabla\Sigma$.
- Magnitude depends on both τ and local field curvature.
- No anomaly should be seen in circular or symmetric trajectories.

9.6 Composite Time and GPS

Satellite navigation systems like GPS are sensitive to relativistic corrections.

- QNG introduces a further correction from τ delay in the propagation of χ.
- For massive, fast-moving systems (e.g., black hole binaries), this may lead to detectable corrections.

9.7 Formal Summary of Composite Time Model

$$T_C = t_R + \underbrace{v \cdot \tau(\chi)}_{\text{Stratonic Lag}} + \underbrace{\delta_\Sigma}_{\text{Fluctuation Coupling}}$$

Where:

- t_R is discrete update time.
- $\tau(\chi)$ is the delay per unit Straton flux.
- δ_Σ represents local field-induced timing variation.

9.8 Testable Predictions

1. Acceleration anomaly in probes with high χ and directed motion.
2. Time dilation deviations in GPS vs GR predictions.
3. Consistent flyby asymmetries depending on entry/exit velocity vectors.
4. Long-term drift of orbiting systems under unmodeled forces.

Chapter 10

Theoretical Implications and Comparison to Other Models

10.1 Motivation for a New Paradigm

Classical approaches to unifying gravity and quantum mechanics struggle with several deep inconsistencies:

- Incompatibility of continuous spacetime with quantum discreteness.
- Singularities in GR (black holes, big bang) indicate breakdowns in geometric models.
- Dark matter and dark energy require exotic particles not yet observed.
- Quantum gravity proposals often lack empirical grounding or predictive clarity.

QNG proposes an alternative: gravity, time, and matter emerge from discrete informational structures, avoiding both singularities and the need for unknown particles.

10.2 Comparison to General Relativity (GR)

- GR assumes smooth spacetime geometry; QNG derives geometry from discrete stability.
- GR uses energy-momentum tensor $T_{\mu\nu}$; QNG replaces it with χ and Σ.
- GR is local and instantaneous; QNG includes time-delayed responses and memory.
- In QNG, gravitational anomalies (e.g. flyby, Pioneer) are predicted structurally.

Emergence of GR:

$$\text{As } \tau \to 0, \quad \Sigma \text{ reduces to standard GR curvature}$$

Thus, GR is a limiting case of QNG, not a contradiction.

10.3 Comparison to Quantum Field Theory (QFT)

- QFT uses fields over continuous spacetime; QNG uses nodes with discrete updates.
- QFT relies on symmetries (gauge invariance, Lorentz invariance); QNG achieves symmetry as emergent, not fundamental.
- QNG avoids ultraviolet divergences due to the lack of infinite frequencies or point particles.

QNG can accommodate QFT-like behavior in stable clusters of nodes, but also explains the origin of quantization itself via fluctuation and persistence rules.

10.4 Comparison to String Theory

- String theory postulates vibrating 1D objects in 10+ dimensions; QNG remains 4D and discrete.
- QNG needs no supersymmetry, branes, or extra spatial dimensions.
- Unlike string theory, QNG is directly testable through gravitational anomalies and memory effects.

QNG Advantage:

QNG does not rely on mathematical elegance alone but links every assumption to empirical observables.

10.5 Comparison to Loop Quantum Gravity (LQG)

- LQG quantizes geometry via spin networks; QNG begins with informational networks.
- Both are background-independent; however, QNG adds dynamics via fluctuation and stability.
- LQG is hard to connect with cosmology; QNG inherently predicts cosmological gradients and lensing.

QNG shares conceptual DNA with LQG but extends it with a richer dynamical substrate.

10.6 Implications for Unification

QNG provides a common framework where:

- Time is emergent from update dynamics.

- Mass and gravity arise from persistence under fluctuation.
- Fields and forces are emergent from phase interactions among nodes.
- Spacetime itself is not fundamental — it's a collective state of the network.

10.7 Philosophical Shift

From: Continuous, predefined spacetime with independent fields.
To: A self-organizing informational system where stability and memory define existence.

This opens new avenues for understanding not just physics, but complexity, emergence, and information itself.

10.8 Schematic Table: QNG vs Other Models

Feature	**GR**	**QFT**	**String**	**QNG**
Spacetime	Continuous	Continuous	10D continuous	Discrete network
Time	Continuous	External parameter	Continuous	Emergent + delayed
Mass	Energy density	Field interaction	Vibrational mode	Temporal stability
Gravity	Geometry	Not included	Closed string	Stability gradient
Dark matter	Unknown particles	Needs WIMPs	Supersymmetric	Memory effect
Testability	High	High	Low	Medium-high

Table 10.1: Comparison of QNG to major physical theories

Chapter 11

Straton Physics: Units, Measurement, and Applications

11.1 Definition of the Straton

In Quantum Node Gravity, the **Straton** (χ) is defined as:

$$\chi = \frac{m}{c}$$

Where:

- m is rest mass [kg],
- c is the speed of light [m/s].

Interpretation:

χ measures the latent informational load that a mass exerts on the spacetime update network — its resistance to transition between nodal states.

Unit: [s·kg/m], a hybrid unit connecting time, mass, and spatial propagation.

11.2 Physical Meaning

- Straton reflects how long it takes a unit of mass to be processed by the spacetime substrate.
- It is the bridge between physical inertia and informational inertia.
- The higher the χ, the more "temporal pressure" it applies on the surrounding Σ-field.

11.3 Theoretical Role in QNG

Fundamental Uses:

- Appears in time-lag expressions: $\Delta L = v \cdot \tau(\chi)$
- Drives acceleration lag: $\vec{a}_{\text{res}} \propto -\chi \cdot \nabla(d\Sigma/dt)$
- Determines rate of integration into new nodal states.

11.4 Experimental Estimation

Let's consider a body of mass $m = 70$ kg. The speed of light $c = 3 \times 10^8$ m/s. Then:

$$\chi = \frac{70}{3 \times 10^8} \approx 2.33 \times 10^{-7} \text{ s·kg/m}$$

This value becomes the base quantity needed for further gravitational, temporal, and delay computations within the QNG framework.

11.5 Straton-Based Equations

Gravitational Update Delay:

$$T_C = t_R + v \cdot \tau(\chi)$$

Memory-Corrected Acceleration:

$$\vec{a} = -\nabla\Sigma + \tau \cdot (\vec{v} \cdot \nabla)\nabla\Sigma$$

Dark Memory Density (DMD):

$$\rho_\chi = \int_{-\infty}^{t} \chi(t') \cdot K(t - t')\, dt'$$

Where $K(t - t')$ is a memory kernel.

11.6 Comparison with Classical Units

11.7 Applications of the Straton

1. Gravitational anomaly prediction

Used to calculate lag-based effects on spacecraft or orbiting bodies.

Quantity	Classical Unit	QNG/Straton-Based Unit
Mass	kg	$\chi \cdot c$
Inertia	kg·m/s^2	$\chi \cdot \nabla\Sigma$
Force	N	Gradient of stability
Acceleration	m/s^2	Time-delayed curvature flux
Energy	J = kg·m^2/s^2	Latent nodal persistence measure

Table 11.1: Physical reinterpretation using Straton framework

2. Detection of hidden mass

Straton flux gradients may map dark matter regions without assuming new particles.

3. Novel unit for informational physics

Could replace or complement Planck-based systems in discrete spacetime models.

4. Engineering implications

If the processing of χ can be simulated in quantum devices, it may lead to new types of computing (temporal resistance logic).

11.8 Future Research Directions

- Developing laboratory analogues to simulate χ interactions.
- Connecting χ to entropy and informational complexity.
- Exploring relativistic corrections to χ in high-energy regimes.
- Searching for cosmological signatures of stratonic lag in early universe dynamics.

Chapter 12

The Composite Time Geometry (CTG): Visualizing the Structure of Time and Causality

12.1 From Linearity to Structure

In classical physics, time is modeled either as:

- **Newtonian:** Absolute and linear.
- **Relativistic:** A geometric coordinate in a continuous spacetime.

In QNG: Time is not an axis, but a multicomponent process:

$$T_C = t_R + v \cdot \tau(\chi)$$

This introduces geometry into time itself — turning it from a background into an emergent, structured quantity.

12.2 Time as Layered Dynamics

We define four distinct layers of temporal structure:

1. **Reference Time** (t_R) — universal network refresh clock.
2. **Relaxation Time** (τ) — system-specific latency caused by χ.
3. **Observed Time** (T_C) — clock time measured macroscopically.
4. **Stratonic Offset** (ΔL) — displacement between state and influence.

$$T_C = t_R + \underbrace{v \cdot \tau(\chi)}_{\text{Stratonic Lag}} + \underbrace{\delta_\Sigma}_{\text{Memory Drift}}$$

12.3 Topological View of Causality

In QNG, events are not aligned along a single timeline. Instead, they are connected through update networks.

Causality: is defined not by coordinate order, but by dependency in state propagation.

Temporal Topology:

Let $\mathcal{N}_i(t)$ be a node state. Then:

$$\mathcal{N}_i(t+1) = \mathcal{U}(\mathcal{N}_i(t), \{\mathcal{N}_j(t)\}, \eta_i)$$

The edges of influence define a temporal flow, forming a directed graph of causation.

12.4 Graphical Visualization (Suggested Diagrams)

- **Diagram 1:** A temporal lattice of nodes updating layer-by-layer, showing delay bubbles based on χ.
- **Diagram 2:** Flow vectors indicating propagation of information in presence of τ.
- **Diagram 3:** Overlap of influence regions, revealing memory "shadows" (potential source of dark matter).

12.5 Composite Light Cone

In relativistic physics, a light cone defines the limit of influence. QNG modifies this through time lag:

$$\text{Causal Cone in QNG:} \quad r \leq c \cdot (t - \tau(\chi))$$

This shrinks the effective causal region depending on mass and informational latency.

12.6 Quantum Interactions over Composite Time

Quantum fluctuations in QNG follow delayed feedback:

$$\phi_i(t+1) = f(\phi_i(t), \phi_j(t), \tau(\chi), \eta)$$

$\rightarrow$ Quantum causality is also dynamic and stratified.

12.7 Implications for Observables

- Events may appear correlated with delay, due to stratonic lag.
- Time measurements in rotating or accelerating frames can deviate from GR predictions.
- Apparent "superluminal" correlations may reflect overlapping memory fields, not true faster-than-light behavior.

12.8 Summary

QNG replaces the timeline with a time-structure:

- Causality becomes nodal and update-driven.
- The observed flow of time depends on informational inertia.
- Temporal curvature and lag lead to testable deviations in astrophysical systems.

This redefinition of time is essential for unifying gravity with quantum dynamics in a discrete, information-centric universe.

Chapter 13

The Dark Sector: Memory, Delay, and Emergent Matter

13.1 Redefining Dark Matter

In standard cosmology, dark matter is:

- Unseen mass causing excess gravitational effects.
- Non-interacting with light (electromagnetism).
- Hypothetical particles (WIMPs, axions) not yet detected.

In QNG: Dark matter is not matter at all.

$$\boxed{\text{Dark Matter} \;\equiv\; \Sigma_{\text{history}} - \Sigma_{\text{now}}}$$

It represents the *persistence of informational curvature* after mass has moved.

13.2 Core Mechanism: Temporal Lag and Memory

When a body of mass m moves, its $\chi = \frac{m}{c}$ takes time τ to be fully integrated into a new node. During that delay, the previous location still maintains gravitational influence.

This creates:

- Gravitational "afterimages"
- Curved spacetime in empty regions
- Apparent hidden mass in observations

13.3 Rewriting the Einstein Equation

In General Relativity:

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\chi}\right)$$

Where:

- $T_{\mu\nu}^{\chi}$ encodes delayed response.
- It behaves observationally as dark matter.
- But is fundamentally memory, not mass.

13.4 Experimental Signatures

Gravitational Lensing:

- Observes $\nabla\Sigma$, not real-time mass.
- Thus, it directly "sees" memory-based gravity.

Galaxy Rotation Curves:

- Outer stars orbit faster than predicted.
- QNG explains this via Σ-lag from historical inner mass.

Bullet Cluster:

- Dark mass offset from luminous mass.
- In QNG: baryons move quickly; Σ lags behind.

13.5 Formal Model:

We define memory-driven curvature:

$$\Sigma(\vec{x}, t) = \int_{-\infty}^{t} K(t - t') \cdot \chi(\vec{x}, t')\, dt'$$

Where $K(t)$ is a memory kernel, such as:

$$K(t) = \frac{1}{\tau} e^{-t/\tau}$$

This leads to a time-smeared version of mass in spacetime.

13.6 Predictive Implications

QNG predicts:

1. Dark matter halos differ by galaxy history.
2. Relaxation leads to slow "fading" of dark matter effects.
3. Gravitational lensing may shift over time.
4. Early universe dynamics encoded memory of high-density fluctuations.

13.7 What Dark Matter is Not

QNG **does not** require:

- Supersymmetry
- WIMPs, axions or sterile neutrinos
- Modified Newtonian Dynamics (MOND)

13.8 What Dark Matter Is (QNG View)

$$\boxed{\text{Dark Matter} = \text{Informational Afterglow of Mass Movement}}$$

No substance, no particles — only memory.

13.9 Cosmological Implications

- Structure formation depends on stability gradients.
- Density fluctuations in the CMB can be reinterpreted as χ-distribution fields.
- Dark matter may vanish in low-χ environments (relaxed regions).

13.10 Conclusion

The QNG model provides a simple but radical view:

- Dark matter is not a mystery particle, but a memory of gravitational influence.
- All known effects can be modeled with χ, τ, and Σ.
- This makes dark matter a testable emergent effect of informational physics.

Chapter 14

Predictive Power: Experimental Tests and Observational Signatures

14.1 Scientific Falsifiability of QNG

A theory is physical only if it can be tested. QNG, unlike many quantum gravity approaches, makes direct and distinct predictions:

- Gravity responds with lag, not instantly.
- Dark matter is a historical memory field, not a particle.
- Acceleration anomalies depend on velocity and nodal flux.

These offer concrete opportunities for falsification and validation.

14.2 Prediction 1: Flyby Anomalies

In spacecraft gravity assist maneuvers (e.g., Galileo, Rosetta), observed speed changes differ slightly from GR predictions.

QNG Explanation: Stratonic lag in gravitational update field causes velocity-dependent residual acceleration.

$$\boxed{\vec{a}_{\text{res}} \approx -\tau(\chi) \cdot (\vec{v} \cdot \nabla)\nabla\Sigma}$$

Expected signature:

- Directional acceleration near perigee
- Magnitude: 10^{-8} to 10^{-10} m/s^2

14.3 Prediction 2: Gravitational Lensing Distortions

Standard Model: Lensing is due to local instantaneous mass.

QNG Prediction: Lensing is based on $\nabla\Sigma$, which can be shifted by memory of past mass locations.

This leads to:

- Offset lensing centers
- Extended or asymmetric arcs
- Time-evolving lensing patterns

14.4 Prediction 3: Rotation Curves Evolution

QNG predicts that:

- Outer galaxy rotation curves depend on accumulated mass history, not just current mass.
- Old galaxies may show "dark matter decay" in halo.

14.5 Prediction 4: Delayed Collapse Structures

Cosmic web structures could show a lag between baryonic clustering and Σ convergence.

Observables:

- Shifted baryon–dark matter centers
- Mismatched velocities in cluster mergers
- Relic lensing in emptied regions

14.6 Prediction 5: Pulsar Timing Variations

In binary pulsars or near dense matter, lag in gravity could produce:

- Oscillatory delays in time-of-arrival (TOA) signals
- Memory-induced gravitational "echoes"

14.7 Test: Modified LIGO Signal Interpretation

Gravitational waves carry Σ updates. If propagation involves $\tau(\chi)$, QNG predicts:

$$\text{Waveform distortion} \propto \text{mass asymmetry} \cdot \tau(\chi)$$

$\rightarrow$ Could explain post-merger ringdown phase variations.

14.8 Lab-Scale Experiments (Future)

- Quantum circuits simulating update lag
- Optical analogues of delayed nodal propagation
- Rotational inertial drift in high-precision torsion balances

14.9 Simulations and Numerical Models

QNG's discrete nature allows efficient simulation:

- Lattice-based gravity dynamics with memory
- Node-based emergent mass systems
- Phase diagram exploration of persistence thresholds

14.10 Conclusion: Scientific Strength of QNG

QNG is not only a theoretical reformulation — it is a testable physical model.

If lag-induced effects are detected, QNG gains ground.If not, its parameters (τ, χ) are constrained.

This positions QNG in a uniquely valuable space: conceptually novel, mathematically rigorous, and scientifically verifiable.

Chapter 15

Integration with Existing Physics: GR, QM, and QFT as Emergent Limits

15.1 Overview: Unification through Emergence

QNG is not a replacement of modern physics, but a unification by grounding classical and quantum behaviors in a deeper informational substrate.

- General Relativity emerges from collective nodal stability gradients.
- Quantum Mechanics arises from stochastic update rules and phase evolution.
- Quantum Field Theory is recovered via coherent phase propagation on structured graphs.

15.2 Recovery of General Relativity (GR)

In QNG:

$$\text{Gravity} = \nabla\Sigma(\vec{x}, t)$$

When Σ is updated continuously and rapidly ($\tau \to 0$), the gradient converges to a classical field:

$$\lim_{\tau \to 0} \nabla\Sigma \longrightarrow R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$$

This produces the Einstein tensor in the smooth limit.
Key Result:

$$G_{\mu\nu} = 8\pi G \cdot T_{\mu\nu} \quad \text{is recovered as a macroscopic approximation}$$

15.3 Recovery of Quantum Mechanics (QM)

Each node possesses a phase variable ϕ_i, and stochastic updates are driven by local rules with noise η_i:

$$\mathcal{N}_i(t+1) = \mathcal{U}(\mathcal{N}_i(t), \{\mathcal{N}_j(t)\}, \eta_i)$$

Under ensemble evolution, this gives rise to:

$$\langle \phi(t) \rangle \longrightarrow \psi(t)$$

Where $\psi(t)$ follows a Schrödinger-like evolution.
Key Mapping:

- Probability amplitude Coherent phase overlap
- Wavefunction collapse Loss of nodal stability
- Measurement Disruption of network coherence

15.4 Recovery of Quantum Field Theory (QFT)

If we treat coherent excitations across many nodes as fields:

$$\Phi(\vec{x}, t) = \sum_i \phi_i(t) \cdot \delta(\vec{x} - \vec{x}_i)$$

Then propagation follows from structured graph dynamics. The field obeys an effective equation:

$$\Box\Phi + V'(\Phi) = 0 \quad \text{(Klein-Gordon-type)}$$

Fermionic behavior: Antisymmetric update rules generate exclusion principles.

15.5 Integration Schema

Emergent Domain	QNG Basis
General Relativity (GR)	$\nabla\Sigma$, $\tau \to 0$
Quantum Mechanics (QM)	ϕ_i, η_i, stochastic updates
Quantum Field Theory (QFT)	Node excitations across graph
Thermodynamics	Persistence + Σ gradients
Inertia	Resistance to nodal phase change

15.6 The Role of the Straton (χ) in All Three Domains

- In GR: χ encodes the effective mass source of Σ
- In QM: χ sets inertia and phase coupling
- In QFT: χ regulates the amplitude and latency of field propagation

15.7 Time Redefined across Frameworks

- GR: Time is geometric — recovered as average nodal update order.
- QM: Time is external parameter — matched by sequential updates.
- QNG: Time is discrete, layered, and intrinsic to the update structure.

15.8 Implications

- All classical results are retained in the proper limit.
- New effects (memory, lag, phase decoherence) emerge only outside classical limits.
- The unification is not imposed — it arises naturally from the architecture.

15.9 Conclusion

QNG achieves a rare balance:

It is novel, but not disruptive. It unifies, but does not override.

It recovers all three major pillars of physics — gravity, quantum mechanics, and fields — as emergent domains of a deeper informational and dynamical structure.

Chapter 16

Mathematical Core of QNG: Axioms, Structures, and Operators

16.1 Axiom Set of Quantum Node Gravity

1. **Discrete Ontology:** Spacetime consists of nodes $\mathcal{N}_i$ with no underlying continuum.
2. **Relational Geometry:** Geometry emerges from relations between nodes, not coordinates.
3. **Temporal Order:** Time is the order of updates, not a geometric axis.
4. **Update-Driven Existence:** An entity exists if and only if it persists through updates.
5. **Stability Selection:** Only configurations with sufficient stability survive.

16.2 Node Structure

Each node $\mathcal{N}_i$ is a triple:

$$\mathcal{N}_i = (V_i,\ \chi_i,\ \phi_i)$$

Where:

- $V_i \in \mathbb{R}^+$: discrete spatial volume
- $\chi_i \in \mathbb{R}^+$: information density (stratonic content)
- $\phi_i \in S^1$: local phase (for coherence and interference)

16.3 Topological Graph Structure

The network is modeled as a graph:

$$\mathcal{G} = (\mathcal{N},\ \mathcal{E})$$

With:

- $\mathcal{N}$: set of nodes
- $\mathcal{E} \subset \mathcal{N} \times \mathcal{N}$: edges defining adjacency

No metric is predefined.

16.4 Temporal Update Rule

Let $t \in \mathbb{N}$ be the discrete time index. Then:

$$\mathcal{N}_i(t+1) = \mathcal{U}\big(\mathcal{N}_i(t), \{\mathcal{N}_j(t)\}_{j \in \mathrm{Adj}(i)}, \eta_i(t)\big)$$

Where:

- $\eta_i(t)$: local fluctuation (stochastic noise)
- $\mathrm{Adj}(i)$: adjacent nodes
- $\mathcal{U}$: update operator

16.5 Fluctuations and Stability

Each node is subject to noise:

$$\eta_i(t) \sim \mathcal{D}_\phi(\sigma), \quad \text{a distribution on phase space}$$

Stability function:

$$\Sigma_i = \Sigma_\chi \cdot \Sigma_{\text{struct}} \cdot \Sigma_{\text{temp}} \cdot \Sigma_\phi$$

Where:

- Σ_χ: memory-based stability
- Σ_{struct}: local graph coherence
- Σ_{temp}: persistence across time
- Σ_ϕ: phase coherence

16.6 Existence Condition

$$\boxed{\Sigma_i \geq \Sigma_{\min} \Rightarrow \text{Node persists}}$$

If not, node is removed from $\mathcal{G}$.

16.7 Emergent Fields

Define a global field:

$$\Sigma(\vec{x}, t) = \sum_i \Sigma_i(t)\ \delta(\vec{x} - \vec{x}_i)$$

Then:

$$\text{Gravity} = \vec{g} = -\nabla\Sigma \qquad \text{Mass} = \Sigma_{\text{temp}}$$

16.8 Memory Kernel (Dark Matter Mechanism)

$$\Sigma(t) = \int_{-\infty}^{t} K(t - t') \cdot \chi(t')\ dt' \quad \text{with} \quad K(t) = \frac{1}{\tau} e^{-t/\tau}$$

This defines historical memory of gravitational influence.

16.9 Composite Time Model (CTM)

Time has two components:

1. Reference Time: t_R, update ordering.
2. Relaxation Delay: τ, processing delay.

Composite time experienced:

$$T_C = t_R + v \cdot \tau = t_R + \Delta L$$

Where ΔL is the stratonic lag distance.

16.10 Stratonic Equation

$$\boxed{\chi = \frac{m}{c}} \quad \text{(Unit of latent reality)}$$

$$\text{Reality Flux} = \frac{d\chi}{dt}$$

16.11 Advanced Operators (Optional)

Define a network Laplacian:

$$\Delta_{\mathcal{G}} f(i) = \sum_{j \in \text{Adj}(i)} (f(j) - f(i))$$

Can be used to define:

- Diffusion-like propagation of information
- Phase synchrony equations
- Coherence thresholds

16.12 Conclusion

This formal structure defines QNG as a fully discrete, graph-theoretic, and information-based gravitational theory with emergent geometry, matter, and temporal experience.

Chapter 17

Cosmological Extensions: QNG and the Large-Scale Universe

17.1 Initial Conditions and the Discrete Genesis

QNG implies that the Universe began not as a singularity in spacetime, but as a minimum-size network of nodes that reached the threshold for existence:

$$\Sigma_i(t = 0) \geq \Sigma_{\min} \quad \forall i$$

This avoids the singularity and instead describes a **Phase Initialization Event (PIE)**, where the first coherent structure emerges due to stochastic fluctuations overcoming a threshold.

17.2 Expansion as Network Proliferation

In QNG, "expansion" is not geometric stretching but:

- Increase in the number of stable nodes
- Increase in relational complexity
- Growth of coherent clusters

$$a(t) \propto |\mathcal{N}(t)|^{1/3}$$

Where $a(t)$ is the emergent scale factor.

17.3 Cosmic Microwave Background (CMB)

Standard Interpretation: Echo of early universe plasma recombination.

QNG View: CMB is the relaxation surface of the initial nodal turbulence. It represents the stabilization of phase structures.

Predictions:

- Slight anisotropies arise from early η_i fluctuation domains
- Power spectrum reflects primordial stratonic coherence zones
- Peak shifts $\rightarrow$ temporal relaxation imprint, not acoustic only

17.4 Structure Formation

Mechanism: Stable clusters of high-Σ nodes attract surrounding unstable zones due to informational gradients.

$$\nabla\Sigma \rightarrow \text{coherent accretion}$$

Dark matter halos: Naturally emerge as memory fields around persistent high-χ regions.

17.5 Large-Scale Filamentary Web

In QNG: Filaments form where:

- Update rules favor directional coherence
- Phase correlations survive over long paths
- τ is locally minimized

$$\text{Filaments} = \text{phase-stable corridors in } \mathcal{G}$$

17.6 Cosmic Acceleration

Standard: Λ-CDM assumes dark energy.

QNG Alternative:

- Apparent acceleration = result of change in nodal update rate
- Phase decoherence $\rightarrow$ reduction in effective χ coupling
- Emergent de-synchronization mimics Λ

17.7 Predictions for Early Universe Anomalies

- Non-Gaussianity: residual correlations from phase initialization
- Cold spots: regions of early nodal instability
- Axis of evil: symmetry remnants in the original node distribution

17.8 Time Arrow and Entropy in QNG

Time emerges from:

$$T = \text{irreversible sequence of updates}$$

Entropy = growth of update-invariant microstates
As structures stabilize, the number of equivalent reconfigurations increases.

$$S(t) \propto \log(\#\text{ of stable configurations})$$

17.9 Inflation vs. Coherence Burst

Instead of rapid inflation, QNG proposes a **Coherence Burst:**

- Sudden increase in Σ_{global}
- Rapid propagation of synchronization phase
- Results in uniformity without violating causality

17.10 Cosmological Timeline in QNG

Epoch	**QNG Interpretation**
PIE (T=0)	First stable node formation
Pre-CMB	Stochastic growth of phase clusters
CMB	Surface of coherence relaxation
Structure era	Accretion via $\nabla\Sigma$
Acceleration era	Global desynchronization phase
Future	Stability decay or saturation

17.11 Conclusion

QNG offers a complete cosmological narrative:

No singularities. No exotic fluids. Just nodes, phases, and stability.

It connects the microscopic structure of spacetime with the macroscopic behavior of the universe using a single coherent framework.

Chapter 18

Applications and Engineering Speculations

18.1 Overview

QNG provides more than a theoretical framework — it offers a new architecture for physical manipulation of space, time, and mass. If the components of the model (stratonic flow, nodal structure, relaxation delays) can be accessed, novel technologies become feasible.

18.2 Straton-Based Mass Engineering

Straton definition:

$$\chi = \frac{m}{c} \quad \Rightarrow \quad m = c \cdot \chi$$

Implication:

- Modifying χ in a nodal region can simulate effective changes in mass without adding physical material.
- This could produce artificial gravitational fields or inertia modifications.

Potential Applications:

- Inertial dampening
- Propulsion without reaction mass
- Mass shielding for impact absorption

18.3 Temporal Manipulation via τ

The relaxation time τ governs how fast a region of space adapts to change.

$$\Delta T = v \cdot \tau \quad \text{(Stratonic Lag)}$$

By engineering local τ, one could:

- Delay or accelerate information transfer
- Create stratonic cloaking — objects not yet "processed" by spacetime
- Build buffers against temporal decoherence

18.4 Stratonic Lensing Systems

Inspired by gravitational lensing:

- Construct artificial stratonic gradients $\nabla\Sigma$
- Refract light or particles by manipulating local stability
- Could act as:
  - Focused communication beams
  - Quantum shielding
  - Directional sensors

18.5 QNG Resonators

Define a QNG cavity:

- Region with fixed phase ϕ_i
- Controlled Σ field
- Induced fluctuations η_i

These could simulate:

- Quantum states without atoms
- Virtual masses
- Temporal coherence states

18.6 Information Transport Networks (QNG-Based)

Instead of transferring particles:

Send a configuration of nodal updates

Implications:

- Massless communication at stratonic-level
- Encoding information in phase synchronization
- Resistant to noise and EM shielding

18.7 QNG-Driven Computation

Define computation as configuration persistence:

$$\text{Logic gates} = \text{Stable nodal transitions}$$

This would allow:

- Topology-based computing
- Fault-resistant architectures
- True spacetime-integrated processors

18.8 Experimental Suggestions

- Measure residual acceleration in satellite flybys vs. τ
- Construct materials with phase-locked structures and monitor Σ behavior
- Probe CMB anomalies with QNG-based simulations

18.9 Cautions and Limits

- Access to χ, ϕ, and Σ may require fundamental breakthroughs
- Control of τ could imply violations of locality
- Ethical implications of manipulating temporal or mass perception

18.10 Conclusion

QNG opens a new domain:

$$\text{Not just physics} \quad \Rightarrow \quad \text{Physics as programmable network}$$

Future generations may treat space and matter the way we treat code today — something to be shaped, compiled, and executed.

Chapter 19

Testing, Falsifiability, and Open Predictions

19.1 Scientific Positioning of QNG

QNG is not a purely philosophical model — it is a physical theory with:

- Axiomatic structure
- Quantifiable variables
- Predictive equations
- Observable deviations from General Relativity (GR)

Its validity depends on its capacity to be **tested**, **falsified**, and **correlated with data**.

19.2 Key Testable Components

19.2.1 1. Stratonic Lag ($\Delta L = v \cdot \tau$)

- Predicts directional residual accelerations
- Appears in spacecraft trajectories (flybys, deep-space probes)
- Observable deviation from GR at low mass/high velocity configurations

19.2.2 2. Stability Field Gradient ($\nabla\Sigma$)

- Predicts gravitational lensing without extra mass
- Simulates dark matter behavior from memory effects
- Confirmable via halo profile discrepancies and lensing offsets

19.2.3 3. Memory Kernel (Historical Σ)

$$\Sigma(\vec{x}, t) = \int_{-\infty}^{t} K(t - t')\, \chi(\vec{x}, t')\, dt'$$

- Predicts time-lagged gravitational response
- Can be simulated in galactic mergers (e.g., Bullet Cluster)

19.3 Experimental Proposals

19.3.1 A. Flyby Anomaly Tracking

- Re-analyze known flybys with QNG correction terms
- Track predicted $\mathbf{a}_{\text{res}} \approx -\tau(\vec{v} \cdot \nabla)\nabla\Sigma$

19.3.2 B. CMB Anomaly Analysis

- Predict cold spots as early nodal instabilities
- Test for residual stratonic patterns in power spectrum

19.3.3 C. Simulated Nodal Networks

- Build QNG-inspired discrete graph simulators
- Introduce stochastic update rules + memory kernels
- Measure emergent behavior matching GR or deviations

19.3.4 D. Laboratory Analogues

- Use condensed matter or photonic systems with delayed response
- Test emergence of curvature-like dynamics from node networks

19.4 Falsifiability Pathways

QNG would be falsified if:

- Flyby and Pioneer data cannot be fit using τ
- Dark matter lensing is observed with no corresponding Σ lag
- Stable nodal simulations cannot replicate GR curvature
- Relaxation equations fail to match cosmological evolution

19.5 Predictions Unique to QNG

1. Residual directional acceleration in low-mass high-velocity bodies
2. Discrepancies between gravitational and electromagnetic centers in merging clusters
3. Temporary "ghost mass" effects in regions with high $\Sigma_{\text{historical}}$
4. Non-Gaussian CMB patterns with anisotropic coherence origin
5. Fluctuation-limited structure formation (early decoherence zones)

19.6 Summary of Test Priorities

Phenomenon	**QNG Signature**	**Dataset**
Flyby Anomaly	Stratonic Lag ΔL	Juno, Galileo, Rosetta
Dark Matter	$\nabla\Sigma$ memory field	Abell 1689, Bullet Cluster
CMB	Phase initialization + τ	WMAP, Planck
Structure Formation	Stability-driven clusters	SDSS, JWST
Space Probes	Temporal relaxation residuals	Pioneer 10/11

19.7 Final Statement

QNG does not challenge observation — it challenges the assumed mechanism. It replaces continuous, instantaneous gravity with a discrete, delayed, and informationally driven process. Its success depends not on aesthetic elegance, but on empirical match.

"If memory shapes gravity, then history shapes the universe."

Appendix A

Definitions, Symbols, Units, and Notations

A.1 Core Concepts and Entities

- **Quantum Node Gravity (QNG)** — A discrete informational framework of gravity where spacetime emerges from the evolution of nodal states.

- **Node** ($\mathcal{N}_i$) — Fundamental unit of reality: contains volume, informational load, and phase.

- **Straton** (χ) — Unit of latent reality, defined as:

$$\chi = \frac{m}{c}$$

- **Stability Function** (Σ) — Composite function determining whether a node exists across updates.

- **Time of Relaxation** (τ) — Delay in the gravitational field adapting to changes.

- **Composite Time ($\mathbf{T_C}$)** — Observable time, resulting from delays in nodal updates.

- **Timpul de Referință ($\mathbf{t_R}$)** — Fundamental system "clock", not directly observable.

- **Gradient of Stability** ($\nabla\Sigma$) — Defines emergent gravitational behavior.

A.2 Symbol Table

Symbol	Definition	Unit / Domain
$\mathcal{N}_i$	Node state	—
V_i	Discrete node volume	m^3 (Planck-scale units)
χ_i	Straton — memory per node	$[kg \cdot s/m]$
ϕ_i	Phase state	$[0, 2\pi]$
Σ_i	Stability function at node i	$[0, 1]$
τ	Relaxation time	$[s]$
T_C	Composite time	$[s]$
t_R	Reference (update) time	$[s]$
$\nabla\Sigma$	Gradient of stability	$[1/m]$
m	Mass (emergent)	$[kg]$
ΔL	Stratonic lag (displacement)	$[m]$
η_i	Fluctuation term (stochastic)	—
$\mathbf{g}$	Emergent gravitational field	$[m/s^2]$

A.3 Constants and Assumptions

- Speed of light: $c = 299,792,458\, m/s$
- Straton minimum: $\chi_{\text{min}} = m_{\text{Planck}}/c$
- Existence condition: $\Sigma_i \geq \Sigma_{\text{min}}$
- All time is modeled discretely: updates indexed by integers

A.4 Notation Conventions

- Vector fields denoted in bold: $\mathbf{g}, \mathbf{v}$
- Scalar fields: Σ, χ, ϕ
- Time indices: $t, t+1, t'$
- Spatial arguments: $\vec{x}, \vec{r}$
- Update functions: $\mathcal{U}()$ for nodal transitions
- Historical integrals: $\int_{-\infty}^{t}$ for memory kernels

A.5 Glossary of Terms

- **Node**: Atomic unit of spacetime in QNG.
- **Fluctuation**: Random variable affecting node state per update.

- **Persistence**: Property of a node that survives over time.
- **Cluster**: Group of stable nodes; interpreted as matter.
- **Phase Correlation**: Determines coherence over distances.
- **Stratonic Cloaking**: A hypothetical region where τ diverges.

A.6 Reference Units

QNG assumes modified Planck units, or normalized systems where:

$$\hbar = c = 1 \quad \text{(in theoretical sections)}$$

Numerical values for engineering approximations are used in practical sections.